PassFrame
tuition

Fraction Operations

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: Convert 7/3 to a mixed number.
Method & answer: 7 ÷ 3 = 2 remainder 1, so 7/3 = 2 1/3.

Standard: Add 1 2/5 + 2 3/5.
Method & answer: Add the whole numbers: 1 + 2 = 3. Add the fractions: 2/5 + 3/5 = 5/5 = 1 whole. Total: 3 + 1 = 4.

Advanced: Divide 3/4 by 2.
Method & answer: Divide the numerator by 2 if it divides evenly, or multiply the denominator by 2: 3/4 ÷ 2 = 3/8.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Convert 9/4 to a mixed number.

2. Which mixed number is equal to 11/3? A) 3 1/3 B) 3 2/3 C) 2 1/3 D) 11/3 cannot be converted

3. Convert 2 3/5 to an improper fraction.

4. Add 1/3 + 1/6.

5. What is 2/3 × 1/4? A) 2/12 B) 3/7 C) 2/7 D) 1/12

6. Divide 2/5 by 2.

7. Add the mixed numbers 1 1/4 + 2 1/4.

8. Select ALL of these that equal 5/2: A) 2 1/2 B) 10/4 C) 2 2/5 D) 15/6

9. Convert 17/5 to a mixed number.

10. Divide 3/7 by 3, giving the answer in simplest form. A) 1/7 B) 3/21 C) 1/21 D) 3/7

11. Subtract: 2 3/4 − 1 1/4.

12. Add 1/2 + 1/5.

13. A recipe needs 1 1/2 cups of flour for one cake, and 2 1/2 cups for a bigger cake. What is the total flour needed for both cakes?

14. Which of these is an improper fraction? A) 3/4 B) 7/5 C) 1 2/3 D) 0/5

15. Multiply 1/2 by 1/4.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Add 1 1/3 + 2 1/6.

2. Subtract 3 1/4 − 1 3/4, giving your answer in its simplest form. A) 1 2/4 B) 1 1/2 C) 2 1/2 D) 2 1/4

3. Multiply 2/3 by 3/5.

6. Subtract 5/6 − 1/4.

7. A painter mixes 2 1/2 litres of blue paint with 1 1/4 litres of white paint. How much paint is there in total?

8. Select ALL of these that equal 3/4 × 2/3: A) 6/12 B) 1/2 C) 2/4 D) 3/8

11. Subtract 4 1/5 − 2 3/5.

12. Multiply 3/8 by 4/5.

13. A tank had 7/8 of its capacity in fuel. After a journey, 1/3 of the tank's capacity was used. What fraction of the tank's capacity is left?

4. Divide 3/4 by 3.

5. What is 4/5 ÷ 2, in simplest form? A) 4/10 B) 2/5 C) 8/5 D) 2/10

9. Add 2/5 + 3/10.

10. Divide 5/8 by 5, in simplest form. A) 5/40 B) 1/8 C) 1/40 D) 5/8

14. What is 1 1/2 + 2/3, as a mixed number? A) 2 1/6 B) 1 5/6 C) 2 2/5 D) 1 1/6

15. Divide 9/10 by 3, then add 1/5 to your answer.

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A jug contains 2 1/3 litres of juice. A second jug contains 1 3/4 litres. How much juice is there in total?

2. What is 3 1/4 − 1 5/6, as a mixed number? A) 1 5/12 B) 1 7/12 C) 2 5/12 D) 1 1/12

3. Convert 29/6 to a mixed number, then subtract 2 whole.

4. Multiply 4/9 by 3/8, simplifying before you multiply.

5. A ribbon is 7/8 m long. It is divided equally into 7 pieces. What length is each piece?

6. Which of these calculations gives an answer greater than 4? A) 1 1/2 + 2 1/4 B) 2 2/3 + 1 1/6 C) 3 3/4 + 1/2 D) 1 7/8 + 1 1/8

7. A length of rope 5/6 m long is cut into 5 equal pieces, and then 2 of those pieces are joined together. What is the length of the joined piece?

8. Select ALL of these fraction multiplications that give an answer of 1/4: A) 1/2 × 1/2 B) 3/4 × 1/3 C) 2/3 × 3/8 D) 1/8 × 3

9. A baker had 3 1/2 kg of flour. She used 1 3/4 kg for bread. How much flour is left?

10. What is 5/6 ÷ 5, then multiplied by 3? A) 1/2 B) 5/18 C) 1/6 D) 15/6

11. Order these from smallest to largest: 1 3/4, 1 2/3, 1 5/6.

12. A 9/10 kg bag of rice is shared equally between 3 people. How much rice does each person get?

13. Select ALL of the following pairs of mixed numbers that have a difference of exactly 1/2: A) 2 3/4 and 2 1/4 B) 3 1/3 and 2 5/6 C) 4 1/8 and 3 5/8 D) 2 1/4 and 1 1/2

14. What is 2/5 of 3/4?

15. A tank is 5/8 full. Water is drained so that only 1/4 of what was there remains. What fraction of the tank is now full?